ZIMBABWE INTEGRATING CAPITAL AND RISK PROGRAMME

('ZICARP')

TS 6 - Determination of the Solvency Capital Requirement

Table of Contents

1. Introduction 3

2. Duties and Accountabilities under ZICARP Framework 5

3. Calculation of the Overall SCR 9

4. Material Risks Covered by the SCR Computation 12

5. TS 6 – A1: Capital Add-Ons under ZICARP 15

6. TS 6 – A2: Principles for recognising risk mitigation techniques in the SCR standard formula 16

7. TS 6 – A3: Glossary 21

1. Introduction

1.1 The main objective of this Technical Specification ('TS') is to outline the basis under which the Solvency Capital Requirement ('SCR') is calculated under the Zimbabwe Integrating Capital and Risk Programme ('ZICARP').

1.2 The SCR will be determined using a standardised formula approach under ZICARP, with the Internal Model approach currently considered out of scope.

1.3 The SCR and Minimum Capital Requirement ('MCR') which are outlined in TS 5 address how required capital should be determined under ZICARP.

1.4 The Standard Formula is a forward-looking risk-based capital measure that covers all material risks of an insurance company. It determines required capital through the application of stress scenarios to the insurer's entire balance sheet under ZICARP.

1.5 Simplified calculations and adaptations for the Zimbabwean environment have been applied.

1.6 The standardised formula has further allowed for the impact of risk mitigation and future management action, as well as the risk reducing impact of diversification benefits.

Application

1.7 This updated TS 6 applies to all insurers that are licensed and supervised by the Insurance and Pensions Commission ('IPEC' or the 'Commission').

Effective Date

1.8 This revised TS 6 Version 3 is effective from 1 January 2023 and should be read in conjunction with Circular 23 of 2021 which implements ZICARP.

2. Duties and Accountabilities under ZICARP Framework

Board of Directors

2.1 The ultimate responsibility for the solvency assessment and management of the insurer rests with the Board of Directors. The Board of Directors of an insurance company must ensure that the insurer always meets all the risk-based capital requirements.

2.2 Specifically, the Board of Directors must ensure the following at all times under ZICARP:

2.2.1 That an insurer holds appropriate level and quality of capital that is commensurate to the risks to which the insurer is exposed;

2.2.2 That an insurer maintains an appropriate level and quality of Own Funds to meet the SCR on a continuous basis;

2.2.3 That there are procedures put in place to continuously monitor the financial health of the insurer, and to identify all possible deteriorations in capital or business conditions of an insurer; and

2.2.4 That IPEC is notified within a reasonable time, normally interpreted as three calendar months of any circumstances that may lead to a deterioration of the financial health of an insurer or of circumstances that may cause an insurer to break any of the risk-based capital requirements under ZICARP.

Actuarial Function

2.3 The Actuarial Function will report to the Board of Directors and provide assurance that the calculations of solvency capital requirements are accurate and compliant with the requirements of the ZICARP framework.

2.4 The Head of the Actuarial Function or the Appointed Actuary shall report to IPEC directly and within one (1) month under the following circumstances:

2.4.1 Where the insurer has breached, or is likely to breach the Absolute Minimum Capital Requirements under ZICARP;

2.4.2 Where the insurer has ceased holding effective reinsurance cover;

2.4.3 Where an insurer or its directors may have contravened the Act or any other law and the contravention may prejudice the interests of the policyholders; and

2.4.4 Any other circumstances established that will compromise the insurer's ability to meet policyholder obligations.

External Auditor

2.5 The External Auditor must audit the financial health of an insurer in accordance with its legal and regulatory obligations. The Auditor must report to the Board of Directors and IPEC any matters identified during the audit process that may cause the insurer not to be financially sound.

Risk Management Function

2.6 The Risk Management Function within an insurer reports to the Board of Directors and must:

2.6.1 Provide specialist analysis and perform risk reviews;

2.6.2 Identify, measure, monitor and manage the risks the insurer faces;

2.6.3 Gain and maintain an aggregated view of the risk profile of the insurer at an enterprise-wide and individual business unit level;

2.6.4 Conduct regular stress testing and scenario analyses, with respect to outliers or matters with low probability but high potential impact; and

2.6.5 Have oversight of ERM and ORSA implementation.

Internal Audit

2.7 The head of the internal audit function shall perform regular independent and objective assessments of key components of the ZICARP framework.

2.8 The head of the internal audit function shall submit a report to the board of the insurer with its material findings and remediations on key components of the ZICARP framework. Such a report will further be furnished to IPEC within thirty days after conducting the audit.audit.

2.9 However, shall there be material issues which prejudice policyholders or threaten viability of the insurer the head of internal audit shall immediately report to the Commission.

Compliance Function

2.10 The head of the compliance function shall have the authority and obligation to promptly inform the chairperson of the board directly in the event of any contravention of the Insurance Act [Chapter 24:07] and any other major non-compliance by a member of management or any staff member for that matter or a material non-compliance by the insurer with the ZICARP framework or an external obligation.

2.11 This shall apply where he or she believes that management is not taking the necessary corrective actions, and a delay would be detrimental to the insurer or its policyholders. The chairperson of the board will be required to report to IPEC any such occurrence and institute an investigation and/or remedial action within seven (7) days.

2.12 Detailed roles and accountabilities under ZICARP are further outlined in TS 1 and Governance and Risk Specifications 2, 3 and 4.

3. Calculation of the Overall SCR

3.1 The SCR refers to the amount of own funds that insurers and reinsurers are required to hold so that they can survive extreme expected losses over a 12-month period at a predetermined confidence level.

3.2 The SCR under ZICARP is calibrated at a 99.5% confidence level, except where simplifications are made.

Risks Covered by the SCR Computation

3.3 The Standardised formula for SCR addresses the following material risk categories:

a) Market Risk

b) Underwriting Risk (both life and non-life); and

c) Operational Risk.

Use of Standardised Formula

3.4 The formula is standardised in the sense that the stress scenarios and required computations are set by the Commission.

3.5 The Commission has further simplified the computations and where relevant made adaptations to suit local conditions.

The Basic Solvency Capital Requirement

3.6 Aggregating the capital requirements for market risk, life underwriting risk and non-life underwriting risk components using correlation parameters prescribed in this specification produces the Basic Solvency Capital Requirement (BSCR). The BSCR is calculated according to the formula:

$$BSCR = \sqrt{\sum_{i,j} CorrBSCR_{i,j} \times SCR_i \times SCR_j}$$

where:

i. SCR_i, SCR_j = The solvency capital requirements which are outlined in TS 6.1, TS 6.2 and TS 6.3

ii. $CorrBSCR_{i,j}$ =A correlation parameter which captures the independence between components i and j as outlined in Table 1.

Diversification Benefits

3.7 To come up with the overall SCR formula, the risk components in **paragraph 3.3** above are aggregated using correlation parameters to allow for diversification benefits. This produces a lower overall SCR than simple linear sum of the capital requirements.

The Correlation Parameters

3.8 A Correlation matrix is applied to reflect the above diversification benefits. It is a standard matrix that has mostly been adopted from Solvency 2.

3.9 The correlation matrix is based on the relationship that has been established between the risk sub-components from other countries. Customised correlation parameters for the Zimbabwe insurance industry will be established over time as more data becomes available.

3.10 $CorrBSCR_{i,j}$ is the value in the i,jth cell of the correlation matrix.

$CorrBSCR$	Market	Life underwriting	Non-life underwriting
Market	1	0.25	0.25
Life underwriting	0.25	1	0
Non-life underwriting	0.25	0	1

Table 1: The Correlation Matrix $CorrBSCR$

Overall SCR Formula

3.11 The overall Solvency Capital Requirement is given by the formula:

$$SCR_{overall} = BSCR + SCR_{op} + SCR_{Capital\ Add\ On}$$

where:

i. $BSCR$ = the basic solvency capital requirement;

ii. SCR_{op} = the operational risk solvency capital requirement as outlined in TS 6.4; and

iii. $SCR_{Capital\ Add\ On}$ = an additional capital add-on that may be imposed from time to time by the Commission as outlined in section **TS 6-A1** of this technical specification.

4. Material Risks Covered by the SCR Computation

4.1 The Own Risk and Solvency Assessment (ORSA) report details how the insurer is managing each of the material quantitative risks that are covered by the SCR computation, as well as qualitative risk management.

4.2 The following is a description of the material quantitative risks covered by the SCR computation;

Market Risk

4.3 Market risk is the risk of loss arising from the impact of movements in market prices on the value of an insurer's assets and liabilities or of loss arising from the default of the insurer's counterparties.

4.4 Market risk arises from a variety of sources. The calculation of the market risk capital requirement under the standardised formula takes account of the following risk components:

4.4.1 **Counterparty/default risk**: The counterparty default risk under ZICARP reflects all possible losses that may arise due to unexpected default or deterioration in credit worthiness of an insurer's counterparties and debtors over the next twelve months.

4.4.2 **Equity risk:** The risk of a change in value caused by deviations of the actual market values of equities and/or income from equities from their expected values.

4.4.3 **Property risk**: The risk of a change in value caused by deviations of the actual property prices.

4.5 TS 6.1 outlines the calculation of the capital requirement for each of the above risk components.

Underwriting Risk

4.6 Underwriting risk is the risk of loss arising from insurance obligations, such as from poor claims experience, expense over-runs and policy lapses. It comprises of life and non-life underwriting risk.

4.7 Life underwriting risk comprises mortality, longevity, disability-morbidity risk, lapse, expense risk, catastrophe risk and retrenchment risk.

4.8 Non-life underwriting risk comprises catastrophe, premium and reserve risk.

4.9 TS 6.2 and 6.3 outline the calculation of life and non-life underwriting risk capital requirement, respectively, for each of the above risk components.

Operational Risk

4.10 Operational Risk is the risk of loss arising from inadequate or failed internal processes, people and systems, or from external events. Operational risk includes compliance, legal risk and management risk, but excludes risks arising from strategic decisions and reputational risk.

4.10.1 **Compliance risk**: The risk of legal or regulatory sanctions resulting in a financial loss, or loss of reputation as a result of an insurer's failure to comply with laws, regulations, rules, related self-regulatory organisation standards and codes of conduct.

4.10.2 **Legal risk**: The possibility that lawsuits, adverse judgements from courts, or contracts that turn out to be unenforceable, disrupt or adversely affect the operations or condition of an insurer.

4.10.3 **Management risk**: Type of business risk. The risk associated with an incompetent management or a management with criminal intentions.

4.11 TS 6.4 outlines the calculation of the capital requirement for operational risk.

Liquidity risk

4.12 Liquidity risk stems from the lack of marketability of an investment that cannot be bought or sold quickly enough to prevent or minimize a loss.

4.13 The SCR computation does not cover liquidity risk. It is assumed that an insurer can experience a liquidity crisis without necessarily experiencing a solvency crisis.

4.14 Liquidity risk under ZICARP is therefore focused on the measures which an insurer should implement to monitor and manage liquidity mismatches between assets and liabilities[1].

[1] Insurers are required to assess, measure and monitor liquidity risk as part of their ORSA process.

5. TS 6 – A1: Capital Add-Ons under ZICARP

5.1 IPEC may require a capital add-on on an insurer under the following exceptional circumstances:

5.1.1 Where there are significant changes in an insurer's risk profile, and the insurer's proposed risk mitigation responses are assessed to be inadequate;

5.1.2 Where an insurer is in breach of any of the requirements under ZICARP Pillars I, II and III, and such breach is viewed as material and has the potential to lead to an understatement of an insurer's SCR requirements calculated using the standard formula; and

5.1.3 Where there are material changes in the operating environment (insurance industry specific events or economywide events) that have the potential to put into jeopardy the financial stability of the insurance industry as a whole.

5.2 Capital add-ons shall only be considered when other measures have failed, are unlikely to succeed or are not feasible. Additionally, the capital add-on shall be proportionate to the material risks arising from the risk management deficiencies in an insurer's operations or the magnitude of the external events which gave rise to the decision of IPEC to set the capital add-on.

Calculation of the capital add-on

5.3 The capital add-on will be calculated as a percentage of SCR ($BSCR + SCR_{op}$). The actual percentage of capital add-on will be determined from time to time and will range from 0% to 100% of the SCR calculated by the insurer.

6. TS 6 – A2: Principles for recognising risk mitigation techniques in the SCR standard formula

Principle 1: Economic effect over legal form

6.1 Risk mitigation techniques should be recognised and treated consistently, regardless of their legal form or accounting treatment, provided that their economic or legal features meet the requirements for such recognition.

6.2 Where risk mitigation techniques are recognised in the SCR calculation, any material new risks shall be identified, quantified and included within the SCR. Where the risk mitigation technique actually increases risk, then the SCR should be increased.

6.3 The calculation of the SCR should recognise risk mitigation techniques in such a way that there is no double counting of mitigation effects.

Principle 2: Legal certainty, effectiveness and enforceability

6.4 The transfer of risk from the undertaking to the third party shall be effective in all circumstances in which the undertaking may wish to rely upon the transfer. Examples of factors which the undertaking shall take into account in assessing whether the transaction effectively transfers risk and the extent of that transfer include:

6.4.1 whether the relevant documentation reflects the economic substance of the transaction;

6.4.2 whether the extent of the risk transfer is clearly defined and beyond dispute;

6.4.3 whether the transaction contains any terms or conditions the fulfilment of which is outside the direct control of the undertaking. Such terms or conditions may include those which:

6.4.4 would allow the third party unilaterally to cancel the transaction, except for the non-payment of monies due from the undertaking to the third party under the contract;

6.4.5 would increase the effective cost of the transaction to the undertaking in response to an increased likelihood of the third party experiencing losses under the transaction;

6.4.6 would oblige the undertaking to alter the risk that had been transferred with the purpose of reducing the likelihood of the third party experiencing losses under the transaction;

6.4.7 would allow for the termination of the transaction due to an increased likelihood of the third party experiencing losses under the transaction;

6.4.8 could prevent the third party from being obliged to pay out in a timely manner any monies due under the transaction; or

6.4.9 could allow the maturity of the transaction to be reduced.

6.5 An undertaking shall also take into account circumstances in which the benefit to the undertaking of the transfer of risk could be undermined. For instance, where the undertaking, with a view to reducing potential or actual losses to third parties, provides support to the transaction, including support beyond its contractual obligations.

6.6 In determining whether there is a transfer of risk, the entire contract shall be considered. Further, where the contract is one of several related contracts the entire chain of contracts, including contracts between third parties, shall be considered in determining whether there is a transfer of risk. In the case of reinsurance, the entire legal relationship between the cedant and reinsurer shall be taken into account in this determination.

6.7 The undertaking shall take all appropriate steps, for example a sufficient legal review, to ensure and confirm the effectiveness and ongoing enforceability of the risk mitigation arrangement and to address related risks. 'Ongoing enforceability' refers to any legal or practical constraint that may impede the undertaking from receiving the expected protection. In the case of financial risk mitigation, the allowance in the SCR of the 'counterparty default risk' derived from the 'financial risk mitigation technique' does not preclude the necessity of satisfying the 'ongoing enforceability'.

6.8 In the case of financial risk mitigation, instruments used to provide the risk mitigation together with the action and steps taken and procedures and policies implemented by the undertaking shall be such as to result in risk mitigation arrangements which are legally effective and enforceable in all jurisdictions relevant to the arrangement and, where appropriate, relevant to the hedged asset or liability.

6.9 Procedures and processes not materialized in already existing financial contracts providing protection at the date of reference of the solvency assessment, shall not be allowed to reduce the calculation of the SCR with the standard formula.

Principle 3: Liquidity and certainty of value

6.10 To be eligible for recognition, the risk mitigation techniques shall be valued in line with the principles laid down for valuation of assets and liabilities, other than technical provisions. This value shall be sufficiently reliable and appropriate to provide certainty as to the risk mitigation achieved.

6.11 Regarding the liquidity of the financial risk mitigation techniques, the following applies:

6.12 The undertaking should have written internal policy regarding the liquidity requirements that financial risk mitigation techniques should meet, according to the objectives of the undertaking's risk management policy;

6.13 Financial risk mitigation techniques considered to reduce the SCR have to meet the liquidity requirements established by the undertaking; and

6.14 The liquidity requirements shall guarantee an appropriate coordination of the liquidity features of the hedged assets or liabilities, the liquidity of the financial risk mitigation technique, and the overall policy of the undertaking regarding liquidity risk management.

Principle 4: Credit quality of the provider of risk mitigation

6.15 Providers of risk mitigation instruments should have an adequate credit quality to guarantee with appropriate certainty that the undertaking will receive the protection in the cases specified by the contracting parties.

6.16 Credit quality should be assessed using objective techniques according to generally accepted practices.

6.17 The assessment of the credit quality of the provider of protection shall be based on a joint and overall assessment of all the features or contracts directly and explicitly linked to the financial risk mitigation technique. This assessment shall be carried out in a prudent manner, in order to avoid any overstatement of the credit quality.

6.18 The correlation between the values of the instruments relied upon for risk mitigation and the credit quality of their provider shall not be unduly adverse, i.e. it should not be materially positive (known in the banking sector as ‘wrong way risk’). As an example, exposures in a company belonging to a group should not be mitigated with Credit Default Swap provided by entities of the same group, since it is very likely that a failure of the group will lead to falls in the value of the exposure and simultaneous downgrade or failure of the provider of protection. This requirement does not refer to the systemic correlation existing between all financial markets as a whole in times of crisis.

Principle 5: Direct, explicit, irrevocable and unconditional features

6.19 Financial risk mitigating techniques can only reduce the capital requirements if:

6.19.1 they provide the undertaking with a direct claim on the protection provider;

6.19.2 they contain an explicit reference to specific exposures or a pool of exposures, so that the extent of the cover is clearly defined and incontrovertible;

6.19.3 they are not subject to any clause, the fulfilment of which is outside the direct control of the undertaking, that would allow the protection provider to unilaterally cancel the cover or that would increase the effective cost of protection as a result of certain developments in the hedged exposure; and

6.19.4 they are not subject to any clause outside the direct control of the undertaking that could prevent the protection provider from its obligation to pay out in a timely manner in the event that a loss occurs on the underlying exposure.

7. TS 6 – A3: Glossary

Capital-at-Risk (CaR): refers to the capital amount that is earmarked to cater for risks.

Confidence level: Statistical measure of the level of certainty regarding an outcome, typically expressed as the probability value $(1 - \alpha)$ associated with α confidence interval.

Linked insurance obligations: means the insurance obligations under an insurance policy are:

a. Not fully guaranteed or partially guaranteed; and
b. Determined solely by reference to the value of particular assets or categories of assets which are specified in the insurance policy and are actually held by or on behalf of the insurer specifically for the purposes of the insurance policy, less deductions specifically provided for in the insurance policy.

Market risk: is the risk of loss to an insurer's assets and liabilities arising from the level or volatility of market prices of financial instruments.

Operational risk: Risk of a change in value caused by the fact that actual losses, incurred for inadequate or failed internal processes, people and systems, or from external events (including legal risk), differ from the expected losses.

Reinsurance: Type of risk mitigation on the basis of an insurance contract between one insurer or pure reinsurer (the reinsurer) and another insurer or pure insurer (the cedant), to indemnify against losses, partially or fully, on one or more contracts issued by the cedant in exchange for a consideration (the premium).

Reserve risk: Relates to incurred claims, i.e. existing claims, (e.g. including IBNR and IBNER), and originates from claim sizes being greater than expected, differences in timing of claims payments from expected and differences in claims frequency from those expected.

Risk margin: A generic term, representing the value of the deviation risk of the actual outcome compared with the best estimate, expressed in terms of a defined risk measure.

Solvency capital requirement: The amount of capital to be held by an insurer to meet the Pillar I requirements under ZICARP.

Technical provision: An insurer's contractual obligations/rights under an insurance contract.

Value-at-Risk: The maximum dollar amount expected to be lost (given normal market conditions) over a given time horizon, at a predetermined confidence level.